

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

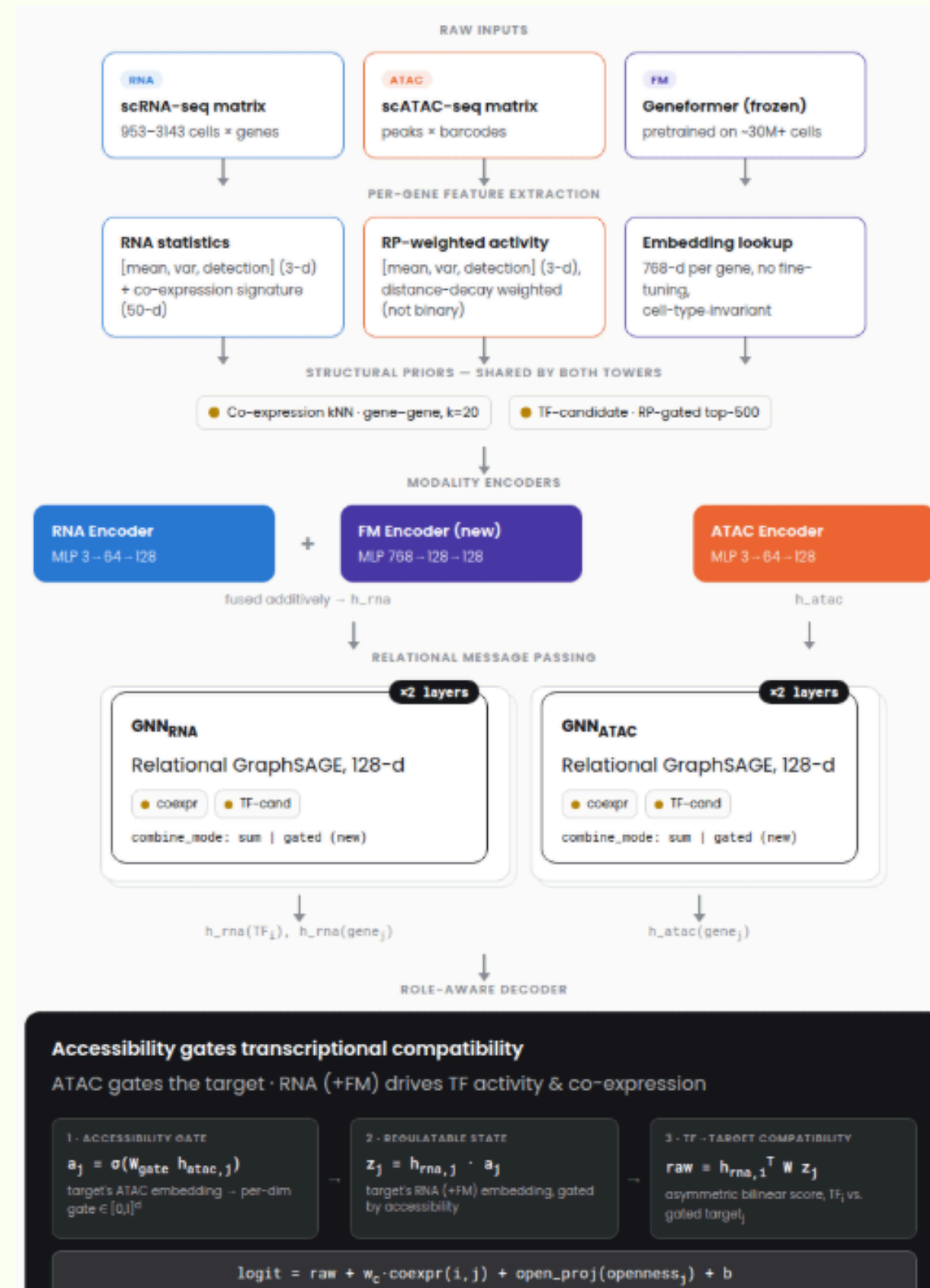
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6th Sept '26 - 12th Sept '26

Weekly Update

Architecture



Architecture

- **Pretrained Geneformer Integration:** Added a frozen 768-d Geneformer embedding tower with an MLP encoder (768 -> 128 -> 128), fused additively into the RNA pathway (parameter count expands from 306K to 430K).
- **Distance-Decayed ATAC Featurization:** Upgraded gene activity from a flat binary window (+/- 100kb TSS) to a continuous, distance-decay weighted Regulatory Potential (RP) model with 3-d statistics ([mean, var, detection]).
- **Enhanced ATAC Encoder:** Replaced the simple linear layer (2 -> 128) with a 2-layer MLP (3 -> 64 -> 128) to process the richer locus descriptors.
- **RP-Gated TF-Candidate Graph:** Shifted candidate edge selection from simple accessibility and co-expression to an RP-gated formulation (RP-gated top-500).
- **Refined Decoder & Edge Scoring:** Upgraded the scalar locus openness offset in the role-aware decoder to an openness projection vector (open_proj).

About the dataset

Metric	K562 (human)
scRNA cells	~950
scATAC cells	~1,100
Transcription factors (TFs)	209
Target genes	~22,900
Localization edges	~1.17 M
Perturbation edges	~173 K
Dual-evidence edges	~27 K

Results against baselines

Method	Loc AUPR	Pert AUPR	Dual AUPR	Dual AUROC
MEvD-GRN (fixed, no embedding)	0.573	0.641	0.881	0.971
MEvD-GRN + Geneformer	0.677	0.800	0.951	0.989
scMultiomeGRN (adapted baseline)	0.901	<i>not reached</i>	<i>not reached</i>	<i>not reached</i>
GRNBoost2	0.535	0.206	0.173	0.525
RegDiffusion	0.546	0.185	0.165	0.490
GMF-GAE	0.526	0.218	0.160	0.516

Ablations

Ablation	Change	Dual AUPR	Dual AUROC
Full model	(reference)	0.881	0.971
no_gnn	remove GNN message passing entirely	0.295	0.720
coexpr_only	GNN sees only the co-expression graph	0.814	0.953
tf_cand_only	GNN sees only the TF-candidate graph	0.785	0.933
rna_only	ATAC features zeroed	0.856	0.962
gated_fusion / concat_fusion / edge_mlp	alternative fusion styles	0.882–0.890	0.970–0.971
gated_relations	learned relation weight instead of fixed sum	0.876	0.969
pert_only	skip localization pretraining entirely	0.441	0.757



THANK YOU!

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